

Li_{3-x}Zr_x(Ho/Lu)_{1-x}Cl₆ Solid Electrolytes Enable Ultrahigh-Loading Solid-State Batteries with a Prelithiated Si Anode

Laidong Zhou, Tongtong Zuo, Chang Li, Qiang Zhang, Jürgen Janek,* and Linda F. Nazar*

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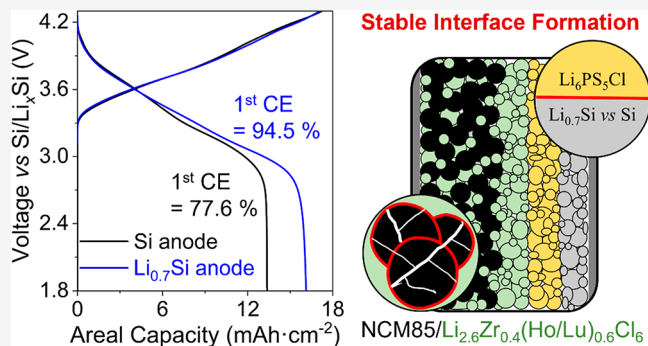


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Supporting Information

ABSTRACT: We report two new families of lithium metal chloride solid electrolytes Li_{3-x}Zr_x(M)_{1-x}Cl₆ (0 ≤ x ≤ 0.8; M = Ho or Lu) with ionic conductivities of up to 1.8 mS cm⁻¹ and a low activation energy of 0.34 eV. Structural elucidation *via* high-resolution neutron diffraction determines the Li ion distribution in trigonal Li₃HoCl₆, orthorhombic-I Li₃LuCl₆, and orthorhombic-II Li_{2.4}Zr_{0.6}(Ho/Lu)_{0.4}Cl₆. The last compound exhibits well-connected Li-ion pathways and abundant Li-ion carriers/vacancies to promote diffusion. All-solid-state batteries with Li_{2.6}Zr_{0.4}(Ho/Lu)_{0.6}Cl₆ solid electrolytes and NCM85 cathodes exhibit stable cycling up to 4.6 V vs Li⁺/Li, which is even preserved up to 4.8 V. Stable cathode interphases are formed for both electrolytes upon cycling to 4.3, 4.6, and 4.8 V cutoff potentials, as identified by a ToF-SIMS analysis. Solid-state cells with a prelithiated Li_{0.7}Si anode exhibit a significantly increased initial coulombic efficiency of 94.5% compared to Si and a high areal capacity of up to 16.3 mAh·cm⁻².



Global decarbonization relies on substantially reducing worldwide energy dependence on fossil fuels and increasing the use of renewable energy resources. Rechargeable batteries are the key component to widely realize the use of clean energy.^{1,2} However, safety concerns about organic liquid electrolytes in conventional lithium-ion batteries may ultimately factor in their application in electric vehicles and grid electricity storage, as electrification becomes more deeply entrenched. All-solid-state batteries (ASSBs) with inorganic solid electrolytes have shown significant potential to provide higher energy density and better safety.^{3–8} They also exhibit good thermal stability at high temperature and good performance at low temperature.

A wide range of inorganic solid electrolytes (SEs) has been investigated, including oxides, sulfides, and chlorides.^{9–11} Among them, sulfides exhibit the highest ionic conductivity (up to 25 mS·cm⁻¹)^{12–14} and can form a quasi-stable interface with a Li metal anode.¹⁵ However, they also display a low oxidation potential of ~2.5 V vs Li⁺/Li.^{16,17} Currently they are suitable as a solid electrolyte combined with a low-voltage cathode such as S/Li₂S,^{18,19} or as a catholyte with a coated high-voltage LiNi_{1-x-y}Co_xMn_yO₂ (NCM)-type cathode. Lithium metal chloride solid electrolytes, on the other hand, are appealing for use in high-voltage cathode composites due to their intrinsic chemical and electrochemical stability with

uncoated NCM cathodes.^{20–22} A number of Li-M-Cl materials have been reported with Li ion conductivities between 0.5 and 2 mS cm⁻¹, including Li₃YCl₆,²³ Li₃InCl₆,²⁴ Li₃Y_{1-x}In_xCl₆,²⁵ Li_xScCl_{3+x},²⁶ Li_{3-x}(Zr/Hf)_x(Y/Er/Yb/In)Cl₆,^{27–30} Li₂(Sc/In)_{2/3}Cl₄,^{20,31} Li₂ZrCl₆,^{32–34} Li_{3-3x}M_{1+x}Cl₆ (M = Tb, Dy, Ho, Y, Er, Tm),³⁵ and Li_{0.5}SmCl_{3.5}.³⁶ The recently reported Li(Ta/Nb)Cl₅X (X = F, Cl, O, OH) family exhibits an even higher ionic conductivity of >10 mS cm⁻¹,^{37,38} while Li_{0.388}Ta_{0.238}La_{0.475}Cl₃ is reported to form a stable interphase with Li metal and stable cycling of Li symmetrical cells at high current density.³⁹ However, these chlorides are based on rare-earth or rare metals and/or are synthesized through high-energy ball milling, which increases cost and may prohibit scalability. To lower the rare (earth) metal content, Li_{3-x}Zr_x(Y/Er/Yb)_{1-x}Cl₆ materials are promising, since they can be synthesized through classical solid-state synthesis, and Zr⁴⁺ substitution importantly leads to an increase in ionic conductivity.^{27,28} The same concept can also be applied to

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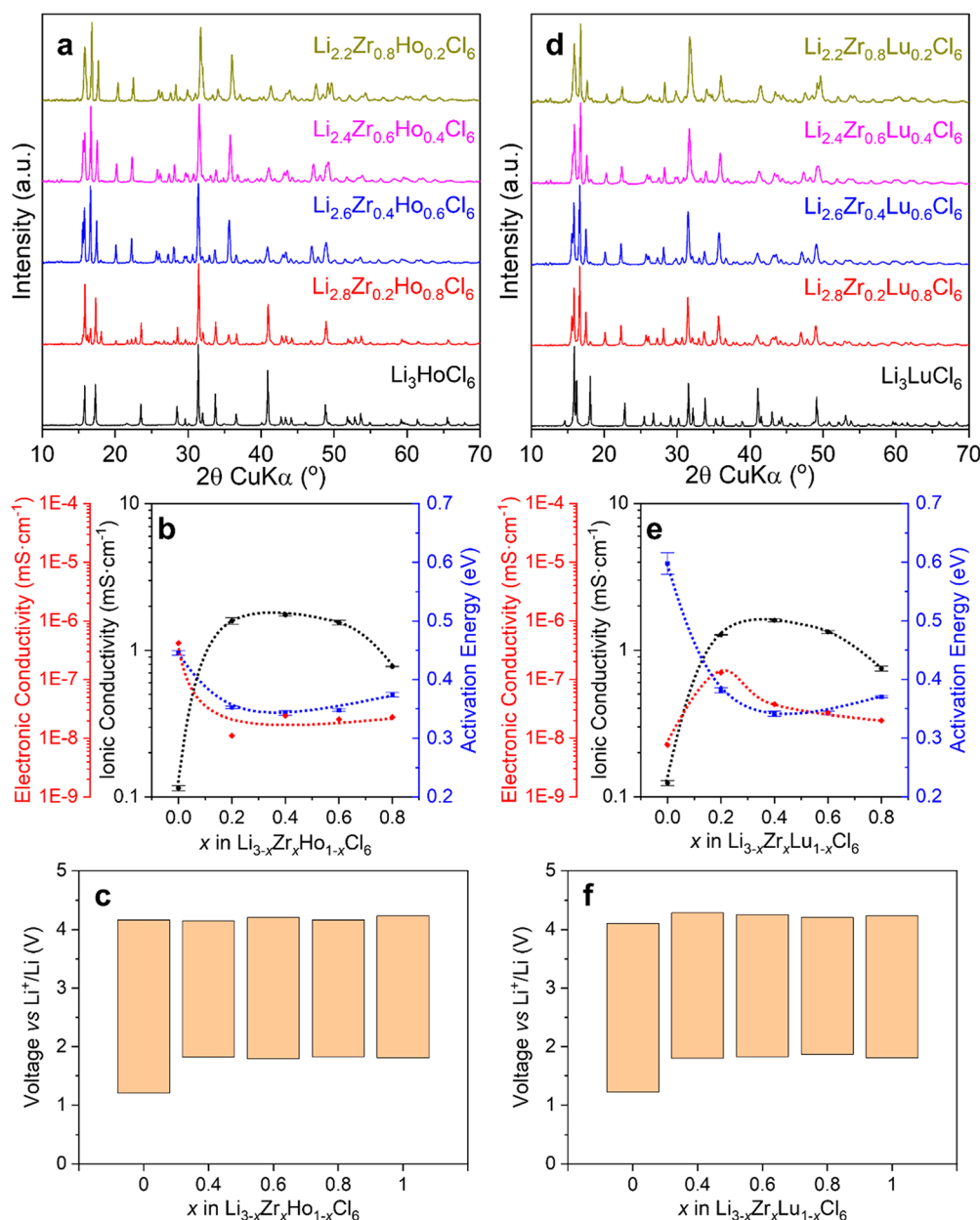


Figure 1. XRD patterns of (a) $\text{Li}_{3-x}\text{Zr}_x\text{Ho}_{1-x}\text{Cl}_6$ and (d) $\text{Li}_{3-x}\text{Zr}_x\text{Lu}_{1-x}\text{Cl}_6$. Room-temperature Li ion conductivities, electronic conductivities, and activation energies of (b) $\text{Li}_{3-x}\text{Zr}_x\text{Ho}_{1-x}\text{Cl}_6$ and (e) $\text{Li}_{3-x}\text{Zr}_x\text{Lu}_{1-x}\text{Cl}_6$ as a function of nominal Zr^{4+} content. Oxidation and reduction potentials of (c) $\text{Li}_{3-x}\text{Zr}_x\text{Ho}_{1-x}\text{Cl}_6$ and (f) $\text{Li}_{3-x}\text{Zr}_x\text{Lu}_{1-x}\text{Cl}_6$ as a function of nominal Zr^{4+} content, measured with cathodic and anodic LSV scans.

other lanthanide-series lithium metal chlorides, which can relieve the dependence on a single rare-earth element. With a large drop in rare-earth element content and the use of multiple elements, this can help lower the cost of chloride SEs and promote their scalable production.⁴⁰ However, our recent work has shown that the chemical and electrochemical stability of Li-M-Cl SEs with NCM cathodes is strongly dependent on the nature of the central metal cation (M).⁴¹ Thus, it is essential to examine the intrinsic stability of other lithium metal chlorides with NCM cathodes.

A challenge of ASSBs is that when a lithium metal anode is employed, most SEs are thermodynamically unstable and degrade or form a solid electrolyte interphase, which may increase the cell impedance. Lithium stripping and plating additionally leads to void and crack formation which promotes

Li dendrite growth.^{42–45} Furthermore, although no/low external pressure ASSBs are needed for practical applications (which requires further exploration⁴⁶), nonetheless NCM cathodes exhibit large volume changes during cycling. The moderately high applied pressure required to maintain contact within the cathode composite is problematic for a soft metal anode. To address this, Meng et al. utilized micrometer-sized silicon for the anode.⁴⁷ Silicon has a high specific capacity exceeding 3500 mAh g^{-1} and exhibits a low electrochemical potential of $\sim 0.3 \text{ V vs Li}^+/\text{Li}$, which will not dramatically lower the cell energy density.⁴⁸ However, due to reactivity between the commonly used argyrodite SE ($\text{Li}_6\text{PS}_5\text{Cl}$) and lithiated Si, formation of the interphase consumes lithium from the cathode and leads to a low discharge capacity (i.e., poor cathode utilization and low first cycle coulombic efficiency of

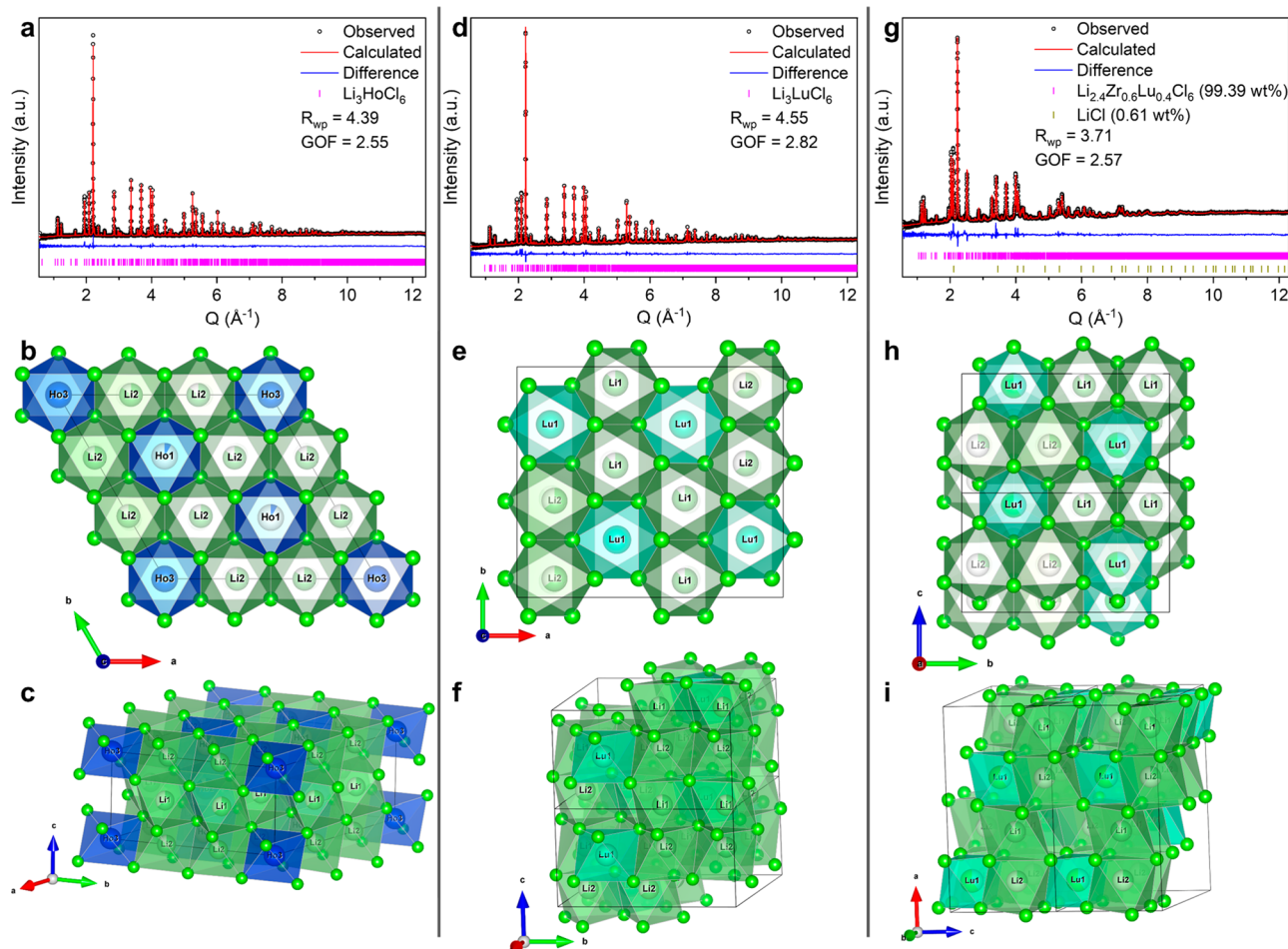


Figure 2. Time-of-flight powder neutron diffraction patterns and the corresponding Rietveld refinement of (a) Li_3HoCl_6 , (d) Li_3LuCl_6 , and (g) $\text{Li}_{2.4}\text{Zr}_{0.6}\text{Lu}_{0.4}\text{Cl}_6$. The corresponding refined structures of Li_3HoCl_6 (b, c), Li_3LuCl_6 (e, f), and $\text{Li}_{2.4}\text{Zr}_{0.6}\text{Lu}_{0.4}\text{Cl}_6$ (h, i).

77.5%). Thus, either overlithiated $\text{Li}_{1+z}\text{Ni}_{1-x-y}\text{Co}_x\text{Mn}_y\text{O}_2$ or prelithiated Si (Li_xSi) is necessary to balance the lithium loss caused by anode side reactions.

Here, we report two new chloride SEs, $\text{Li}_{3-x}\text{Zr}_x\text{Ho}_{1-x}\text{Cl}_6$ and $\text{Li}_{3-x}\text{Zr}_x\text{Lu}_{1-x}\text{Cl}_6$, with high ionic conductivities of $\sim 1.8 \text{ mS cm}^{-1}$ and low activation energies of $\sim 0.34 \text{ eV}$. Using high-resolution powder neutron diffraction and X-ray diffraction, we identify the trigonal (Li_3HoCl_6) to orthorhombic-II ($\text{Li}_{2.4}\text{Zr}_{0.6}\text{Lu}_{0.4}\text{Cl}_6$) phase transition upon Zr^{4+} substitution and local structural rearrangement change of orthorhombic-I Li_3LuCl_6 to orthorhombic-II $\text{Li}_{2.4}\text{Zr}_{0.6}\text{Lu}_{0.4}\text{Cl}_6$. ASSBs comprised of $\text{Li}_{2.6}\text{Zr}_{0.4}(\text{Ho/Lu})_{0.6}\text{Cl}_6$ SEs paired with high-Ni $\text{LiNi}_{0.85}\text{Co}_{0.1}\text{Mn}_{0.05}\text{O}_2$ (NCM85) show stable cycling at cutoff potentials of both 4.3 and 4.6 V vs Li^+/Li . Even upon cycling to 4.8 V, cells exhibit relatively stable long-term cycling. A time-of-flight secondary ion mass spectrometry (ToF-SIMS) analysis shows that the NCM85/chloride interface degrades at different rates with different cutoff potentials but stabilizes with cycling to prevent further degradation of the electrolytes. NCM85 ASSBs with $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ as the SE and Si or $\text{Li}_{0.7}\text{Si}$ as the anode demonstrate stable long-term cycling at a high areal capacity of 16 mAh cm^{-2} . The use of $\text{Li}_{0.7}\text{Si}$ compared to Si significantly increases the initial coulombic efficiency and cathode utilization.

Figure 1a,d shows the powder X-ray diffraction patterns of as-synthesized phase-pure $\text{Li}_{3-x}\text{Zr}_x(\text{Ho/Lu})_{1-x}\text{Cl}_6$ ($0 \leq x \leq 0.8$). Li_3HoCl_6 adopts a trigonal structure ($P\bar{3}m$), but with

increasing Zr^{4+} content, $\text{Li}_{3-x}\text{Zr}_x\text{Ho}_{1-x}\text{Cl}_6$ ($0.2 < x \leq 0.8$) transitions to an orthorhombic-II structure ($Pnma$). Li_3LuCl_6 exhibits an orthorhombic-I structure ($Pnma$), but with increasing Zr^{4+} substitution, $\text{Li}_{3-x}\text{Zr}_x\text{Lu}_{1-x}\text{Cl}_6$ ($0.2 \leq x \leq 0.8$) converts to the same orthorhombic-II framework with tilted $(\text{Lu/Zr})\text{Cl}_6$ octahedra. This structural transition is similar to that reported for $\text{Li}_{3-x}\text{Zr}_x(\text{Y/Er})_{1-x}\text{Cl}_6$.²⁷ The SEM images of $\text{Li}_3(\text{Ho/Lu})\text{Cl}_6$ and $\text{Li}_{2.6}(\text{Ho/Lu})_{0.6}\text{Zr}_{0.4}\text{Cl}_6$ (Figure S1) reveal micrometer-sized particles. The room-temperature ionic conductivity and activation energy of $\text{Li}_{3-x}\text{Zr}_x(\text{Ho/Lu})_{1-x}\text{Cl}_6$ ($0 \leq x \leq 0.8$) are shown in Figure 1b,e, and the corresponding variable-temperature Nyquist plots and Arrhenius plots are shown in Figures S2–S5. Both parent phases, Li_3HoCl_6 and Li_3LuCl_6 , exhibit a relatively low ionic conductivity of $\sim 0.1 \text{ mS cm}^{-1}$. With minor Zr^{4+} substitution that triggers formation of the orthorhombic-II phase, all solid solutions, $\text{Li}_{3-x}\text{Zr}_x(\text{Ho/Lu})_{1-x}\text{Cl}_6$ ($0.2 \leq x \leq 0.6$), exhibit a much higher ionic conductivity (up to 1.8 mS cm^{-1}) and a relatively low activation energy of $\sim 0.34 \text{ eV}$. The electronic conductivities (Figure 1b,e) of $\text{Li}_{3-x}\text{Zr}_x(\text{Ho/Lu})_{1-x}\text{Cl}_6$ ($0 \leq x \leq 0.8$) measured with direct current polarization (Figures S6 and S7) are all extremely low ($< 10^{-7} \text{ mS cm}^{-1}$), which is beneficial for kinetically stable operation at a high potential.²⁰ The electronic conductivity from DC polarization studies is determined by the steady-state current, which will not be reached if the polarization time is too short. The value can also depend on the technique used for measurement, making it

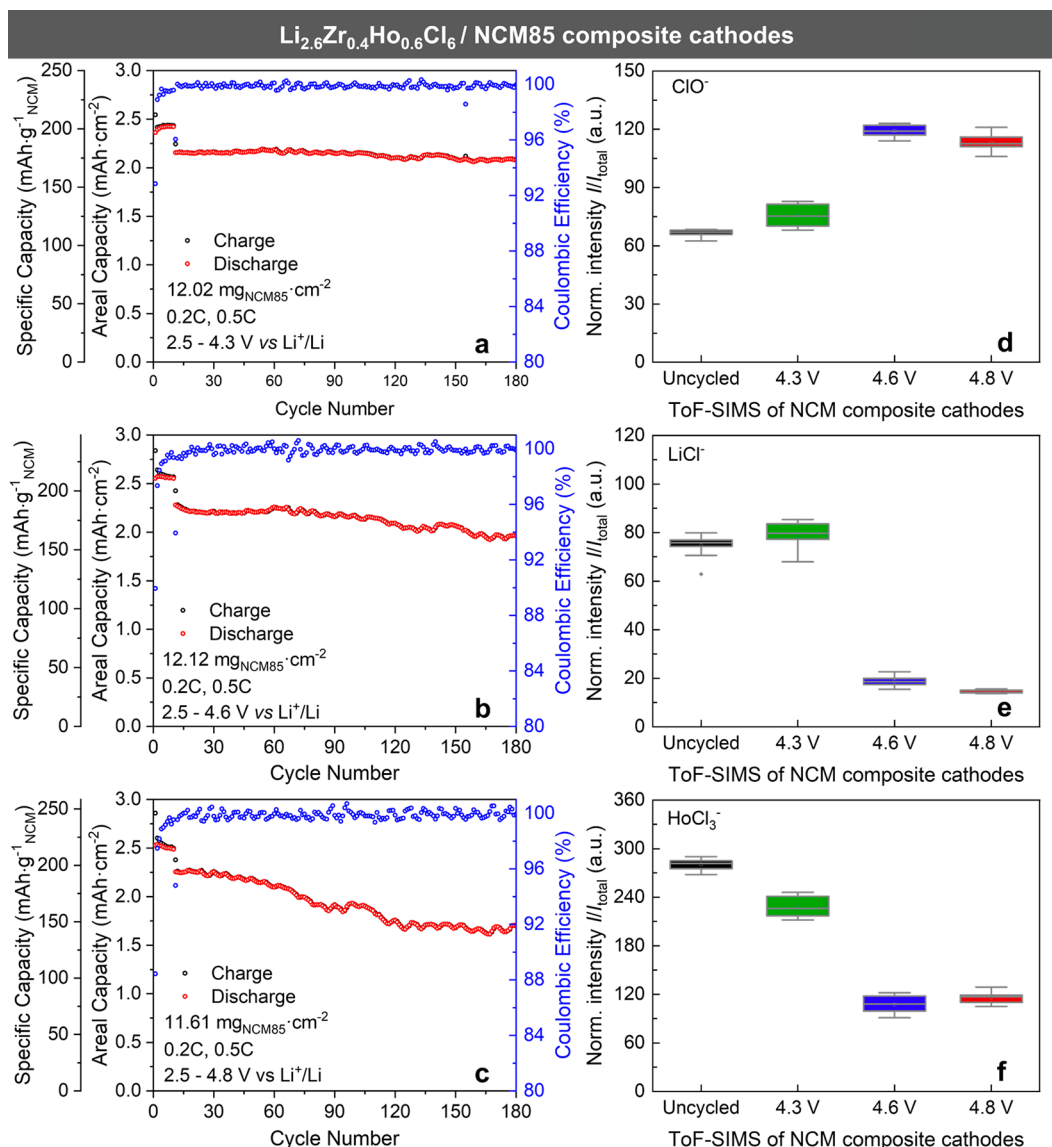


Figure 3. Cycling performance of the bare NCM85 ASSBs with $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Ho}_{0.6}\text{Cl}_6$ as catholyte at cutoff potentials of 4.3 V (a), 4.6 V (b), and 4.8 V vs Li^+/Li (c). The slight oscillations of the cycling data are due to the room-temperature variations. ToF-SIMS surface analysis of NCM85 cathode composites uncycled and cycled for 180 cycles with different cutoff potentials with $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Ho}_{0.6}\text{Cl}_6$ as catholyte (d–f). Shown are box plots for ClO^- , LiCl^- , and HoCl_3^- fragment signal intensities normalized to the respective total ion signal, containing 5 measurements per sample.

difficult to compare literature data.⁴⁹ Table S1 gives electronic conductivity data for representative chloride SEs, all measured using exactly the same protocol (Li_3InCl_6 , Li_3YCl_6 , $\text{Li}_{2.5}\text{Y}_{0.5}\text{Zr}_{0.5}\text{Cl}_6$, ball-milled Li_2ZrCl_6 , $\text{Li}_2\text{Sc}_{0.667}\text{Cl}_4$, and $\text{Li}_2\text{Sc}_{0.333}\text{In}_{0.333}\text{Cl}_4$). They all exhibit similar very low electronic conductivities; in contrast, Li_3HoCl_6 and Li_3LuCl_6 show rather different electronic partial conductivities (Figures S6–S8), while the Zr-substituted materials again show fairly similar electronic behavior (Figure 1b,e). In line with our previous report on the role of the transition-metal ion(s) for the electronic and stability properties of halide electrolytes,⁴¹ it appears that different lanthanide metal ions can trigger quite different electronic properties and exhibit different chemical reactivity with O_2 released from NCM85 at a high potential that may be decisive for interphase growth under oxidizing conditions. We highlight that the electronic properties of solid electrolytes remain an important subject for future studies. The

stability voltage range of $\text{Li}_{3-x}\text{Zr}_x(\text{Ho/Lu})_{1-x}\text{Cl}_6$ ($0 \leq x \leq 0.8$) measured with linear sweep voltammetry (LSV) (Figures S9 and S10) demonstrates a rather similar oxidation potential of ~ 4.2 V vs Li^+/Li determined by the Cl^- anion framework, i.e., the occupied Cl (2p) states in the band structure. The reduction potential is ~ 1.8 V vs Li^+/Li for Zr-substituted materials as dictated by Zr^{4+} reduction and ~ 1.2 V vs Li^+/Li for $\text{Li}_3(\text{Ho/Lu})\text{Cl}_6$ as determined by $\text{Ho}^{3+}/\text{Lu}^{3+}$ reduction.

The crystal structures of trigonal Li_3HoCl_6 , orthorhombic-I Li_3LuCl_6 , and orthorhombic-II $\text{Li}_{2.4}\text{Zr}_{0.6}(\text{Lu/Ho})_{0.4}\text{Cl}_6$ were solved by Rietveld refinement of time-of-flight powder neutron diffraction patterns (Figure 2a,d,g and Figure S11a). The corresponding crystallographic details are summarized in Tables S2–S5. Trigonal Li_3HoCl_6 exhibits Li/Ho honeycomb ordered layers (Figure 2b), and within the layer, Li ions can diffuse by passing through low-energy tetrahedral voids. Along the c axis (Figure 2c), the face-sharing Li octahedra provide a

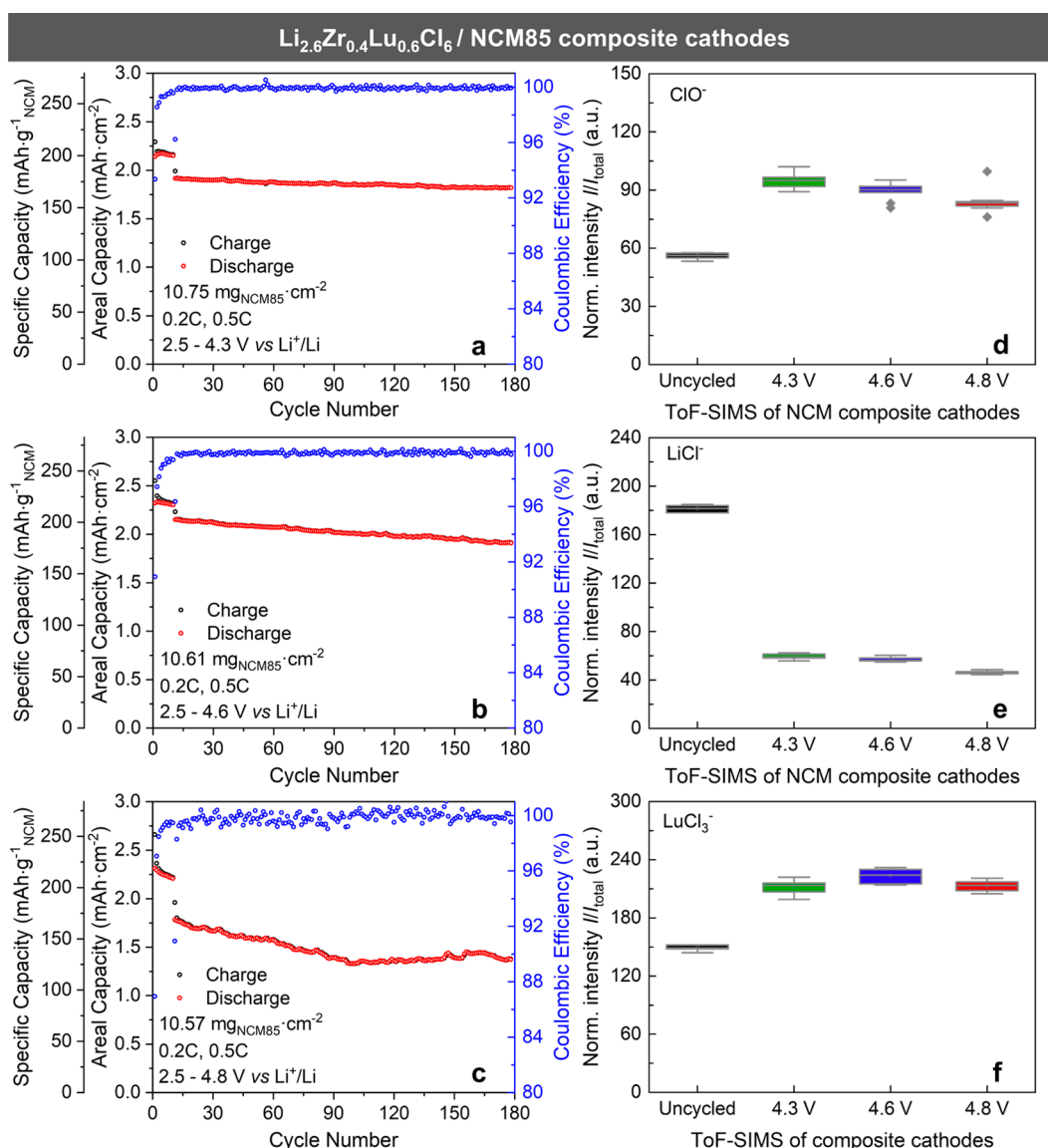


Figure 4. Cycling performance of the bare NCM85 ASSBs with $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ as catholyte at cutoff potentials of 4.3 V (a), 4.6 V (b), and 4.8 V vs Li^+/Li (c). The slight oscillations of the cycling data are due to the room temperature variations. ToF-SIMS surface analysis of NCM85 cathode composites uncycled and cycled for 180 cycles with different cutoff potentials with $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ as catholyte (d–f). Shown are box plots for ClO^- , LiCl^- , and LuCl_3^- fragment signal intensities normalized to the respective total ion signal, containing 5 measurements per sample.

direct Li ion migration pathway. For orthorhombic-I Li_3LuCl_6 , similar Li/Lu honeycomb ordered layers (Figure 2e) with a different LuCl_6 arrangement exist, while Li ions can diffuse within the layer as described above. The preferred c axis diffusion pathway again consists of face-sharing LiCl_6 octahedra (Figure 2f). For orthorhombic-II $\text{Li}_{2.4}\text{Zr}_{0.6}(\text{Lu}/\text{Ho})_{0.4}\text{Cl}_6$, a similar Li/(Lu/Ho/Zr) honeycomb ordered layer (Figure 2h and Figure S11b) with a different arrangement provides a Li ion migration path in the b,c plane. The face-sharing and edge-sharing LiCl_6 octahedra form a zigzag chain along the a axis for Li ion migration (Figure 2i and Figure S11c). In these three structures, all the Li sites are not fully occupied and offer sufficient Li vacancies along the Li ion diffusion pathway, which leads to high Li ion conductivity.⁵⁰ This was further confirmed with BVSE calculations (Figures S12 and S13).^{51,52}

Taking the ionic conductivity and the electrochemical stability window into consideration, we chose $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ and $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Ho}_{0.6}\text{Cl}_6$ as SEs to investigate their electrochemical performance in ASSBs. Using an uncoated NCM85 cathode active material and (initially) In/LiIn as the negative electrode, ASSBs were fabricated (see the Supporting Information for details). A thin layer of argyrodite $\text{Li}_6\text{PS}_5\text{Cl}$ was used to separate the chloride SE from the In/LiIn anode to prevent its reduction.⁵³ The cathode composites were composed of 80 wt % NCM85/20 wt % chloride SE with a high cathode loading of $\sim 11 \text{ mg}_{\text{NCM85}} \text{ cm}^{-2}$. Bare NCM85 was used as received with $D_{50} \approx 5 \mu\text{m}$, as shown in Figure S14a. ASSBs were cycled with different cutoff potentials of 4.3, 4.6, and 4.8 V vs Li^+/Li at room temperature and under 250 MPa pressure, while the first 10 cycles were conducted at 0.2C to form a stable $\text{Li}_6\text{PS}_5\text{Cl}$ -In/LiIn interface. Figure 3a,b shows that the $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ ASSBs exhibit excellent cycling stability with 4.3 and 4.6 V

cutoff potentials. At a 0.2C rate, high discharge capacities of 203 and 213 mAh g⁻¹ were obtained (Figure S15a,b) on cycling to 4.3 and 4.6 V. Even at 0.5C, capacities of 180 and 188 mAh g⁻¹ are maintained (Figure S15a,b) and the cells exhibit 96.8% and 85.6% of the initial capacity after 180 cycles, with respective cutoff potentials of 4.3 and 4.6 V. With a higher cutoff potential of 4.8 V, the ASSB exhibits higher discharge capacities of 218 mAh g⁻¹ at 0.2C and 195 mAh g⁻¹ at 0.5C (Figure S15c). At such a high cutoff potential, the cell exhibits relatively fast capacity fading initially but stabilizes after 100 cycles (Figure 3c), which indicates that a stable interphase has been created between NCM85 and Li_{2.6}Zr_{0.4}Ho_{0.6}Cl₆. After 180 cycles, the cell maintains 75.0% capacity.

ASSBs with Li_{2.6}Zr_{0.4}Lu_{0.6}Cl₆ exhibit the same excellent electrochemical performance. ASSBs exhibit excellent cycling stability and a high discharge capacity of 202 and 220 mAh g⁻¹ at 0.2C, and 202 mAh g⁻¹ at 0.5C using 4.3 and 4.6 V cutoff potentials, respectively (Figure 4a,b and Figure S15d,e). Correspondingly, the cells maintain 95.9% and 88.9% capacity after 180 cycles. At a higher cutoff potential of 4.8 V, the cells deliver 219 mAh g⁻¹ at 0.2C and 168 mAh g⁻¹ at 0.5C as shown in Figure S15f. They similarly demonstrate capacity fading over the initial 100 cycles, which then stabilizes to maintain a 75% capacity over 180 cycles (Figure 4c). As in the case of the Ho SE, this demonstrates formation of a stable interphase. EIS measurements taken before and after 180 cycles (Figure S16) indicate an increase in the cell internal resistance. The bulk solid electrolyte separator layer resistance decreases slightly after cycling for all cells (Figure S16) due to consolidation during long-term cycling.⁵⁵ No increase in the interfacial resistance between Li_{2.6}Zr_{0.4}(Ho/Lu)_{0.6}Cl₆ and Li₆PS₅Cl was noted, indicating their chemical compatibility. We note that other work suggests incompatibility between chloride and sulfide solid electrolytes, but this contradicts our results.^{54,55} This suggests that the compatibility between chloride and sulfide solid electrolytes could be material dependent. The semicircles in the Nyquist plot are mainly attributed to Li ion transport in the cathode composite and to the NCM85–chloride solid electrolyte interphase, according to the model applied by Zhang et al.⁵⁶ The latter resistance clearly dominates the overall response, which corresponds to formation of a thicker Li ion blocking interphase at higher cutoff potentials that drives faster capacity fading. Nevertheless, both Li_{2.6}Zr_{0.4}Ho_{0.6}Cl₆ and Li_{2.6}Zr_{0.4}Lu_{0.6}Cl₆ still show superior electrochemical stability with NCM85 compared to any previously reported mixed-metal system, i.e., Li_{3-x}Zr_x(Y/Er/Yb)_{1-x}Cl₆.^{27–29,41}

Interphase growth in NCM85–Li_{2.6}Zr_{0.4}(Ho/Lu)_{0.6}Cl₆ was further probed by ToF-SIMS.⁵⁷ Uncycled NCM85–Li_{2.6}Zr_{0.4}(Ho/Lu)_{0.6}Cl₆ cathode composites and 180-cycle cathode composites with different cutoff potentials of 4.3, 4.6, and 4.8 V were examined. Analysis was performed on the top surfaces of the cathode composites that were oriented toward the current collector. Based on previous studies,⁴¹ chemical degradation of the lithium metal chloride solid electrolyte with oxygen released from NCM85 at high potentials (>4.2 V) consumes Zr⁴⁺ and Ho³⁺/Lu³⁺ to form ClO⁻-related decomposition products (potentially ZrO₂ and HoClO/LuClO).^{41,58,59} This reaction generally results in the observation of ClO⁻ fragments in the ToF-SIMS data and a decrease of HoCl₃⁻/LuCl₃⁻ fragment signal intensities. Conversely, the electrochemical oxidation of lithium metal chloride solid electrolytes will lead to the formation of Cl₂ and

HoCl₃/LuCl₃ decomposition products,⁶⁰ which will decrease the LiCl⁻ fragment signal intensities and increase those from HoCl₃⁻/LuCl₃⁻ fragments. Thus, the ClO⁻ fragment signal intensity can be regarded as the major indicator of oxygen-related chemical decomposition, whereas LiCl⁻ fragments are indicative of electrochemical decomposition. The HoCl₃⁻/LuCl₃⁻ signal intensity changes can reveal the relative degree of the chemical/electrochemical degradation. To avoid topographic interference and ensure reproducibility, we measured 10 mass spectra in different areas on each sample. The surface analyses of the ToF-SIMS measurements are displayed in the box plots, which show the ClO⁻, LiCl⁻, and (Ho/Lu)Cl₃⁻ signal intensities normalized to the respective total ion intensity (Figures 3d–f and 4d–f). The corresponding mass spectra of these fragments are shown in Figure S17. In uncycled cathode composites, the ClO⁻ signal originates from the chloride solid electrolyte itself (Figures 3d and 4d and Figure S17a,d). As trace moisture is present in the precursors, (Lu/Ho)ClO can form after heat treatment through HoCl₃/LuCl₃ hydrolysis.⁶¹ For Li_{2.6}Zr_{0.4}Ho_{0.6}Cl₆, the LiCl⁻ fragment signal intensity (Figure 3e and Figure S17b) remains nearly unchanged at a 4.3 V cutoff potential, which indicates its good electrochemical stability with NCM85. The ClO⁻ signal intensity (Figure 3d and Figure S17a) increases slightly, which indicates a minor degree of oxygen-induced chemical degradation. Therefore, the HoCl₃⁻ signal intensity (Figure 3f and Figure S17c) decreases slightly. These results demonstrate very good interfacial stability at a 4.3 V cutoff potential, in agreement with the anodic stability window measured by LSV (Figure 1c).

However, when the cutoff potential increases to 4.6 and 4.8 V, the ClO⁻ signal intensity almost doubles compared to the uncycled value (Figure 3d and Figure S17a) and the HoCl₃⁻ signal intensity (Figure 3f and Figure S17c) accordingly decreases significantly. This indicates the formation of ClO⁻-related species. The LiCl⁻ signal intensity (Figure 3e and Figure S17b) also decreases strongly, which indicates significant electrochemical degradation of Li_{2.6}Zr_{0.4}Ho_{0.6}Cl₆ at the higher potentials. Thus, not surprisingly, both chemical and electrochemical degradation occur when the cutoff potential is much higher than the thermodynamic oxidation limit. Importantly, though, all fragment signal intensities are about the same at both 4.6 and 4.8 V (Figure 3d,e and S17b,c). This indicates that chemical and electrochemical degradation leads to initial growth of an interphase (probably HoClO, ZrO₂, and LiCl), which prevents further degradation of the electrolyte and acts as a stable interface.

In contrast, for Li_{2.6}Zr_{0.4}Lu_{0.6}Cl₆, all three ClO⁻, LiCl⁻, and LuCl₃⁻ signal intensities (Figure 4d,e and Figure S17d,e) remain nearly the same after cycling at different cutoff potentials. This highlights a rather stable interface buildup already at 4.3 V. The ClO⁻ signal intensity (Figure 4d and Figure S17d) increases by ~50% (much less than that of Li_{2.6}Zr_{0.4}Ho_{0.6}Cl₆ at high cutoff potentials of 4.6 and 4.8 V), suggesting relatively less oxygen-involving degradation. Nonetheless, even at 4.3 V, a significant increase in the ClO⁻ signal (Figure 4d and Figure S17d) is already observed for Li_{2.6}Zr_{0.4}Lu_{0.6}Cl₆. This result demonstrates that Li_{2.6}Zr_{0.4}Lu_{0.6}Cl₆ is more prone to react with released oxygen as low as 4.3 V, but the oxygen-based degradation products remain stable at higher voltages. Around the same degree of LiCl⁻ intensity drop (Figure 4e and Figure S17e) is observed for both Li_{2.6}Zr_{0.4}Ho_{0.6}Cl₆ and Li_{2.6}Zr_{0.4}Lu_{0.6}Cl₆, highlighting

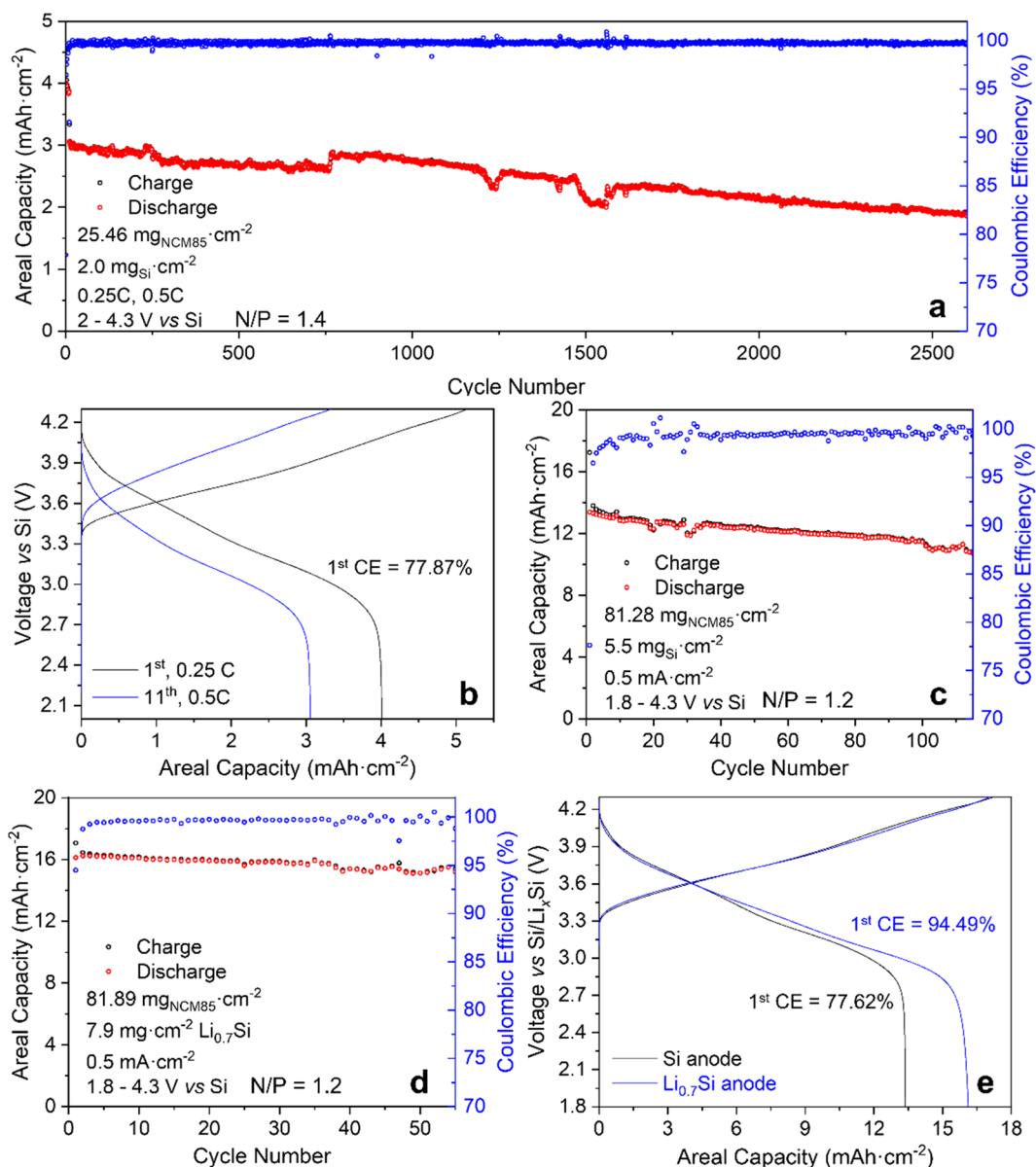


Figure 5. (a) Cycling performance of the bare NCM85 ASSBs with $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ as catholyte and a Si anode in the potential range 2–4.3 V vs Li^+/Li and corresponding voltage profile (b). Cycling performance of ultrahigh-loading ASSBs with Si anode (c) and $\text{Li}_{0.7}\text{Si}$ anode (d) and corresponding 1st cycle voltage profile. The slight oscillations of the cycling data are due to the room-temperature variations.

their similar degrees of electrochemical degradation. The LuCl_3^- signal intensity increases after cycling, also suggesting electrochemical degradation. We note the contradictory trends of the HoCl_3^- and LuCl_3^- signals, which can be attributed to their different relative fractions of chemical and electrochemical degradation processes. The decrease in HoCl_3^- fragment intensity indicates a higher degree of chemical degradation over electrochemical degradation. In contrast, the increasing LuCl_3^- fragment intensity indicates a greater electrochemical degradation. We note that the oxygen-related chemical degradation occurs above 4.15 V for a high-Ni NCM cathode⁶² whereas electrochemical degradation occurs above 4.2 V (the thermodynamic oxidation potential). Thus, both chemical and electrochemical degradation occur simultaneously on charge, and the resulting interphase formation strongly depends on subtle kinetic factors.

Overall, the ToF-SIMS analyses highlight that $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Ho}_{0.6}\text{Cl}_6$ is more chemically reactive with the oxygen

species released from NCM85 at high potential, and $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ is electrochemically less stable. Although the ToF-SIMS data show the same interphase formation with different cutoff potentials, at higher cutoff potentials the cells exhibit much faster capacity fading, especially for $\text{Li}_{2.6}\text{Zr}_{0.4}\text{Lu}_{0.6}\text{Cl}_6$ —and the EIS data show an increasing NCM85–chloride solid electrolyte interface resistance. This could be due to the structural instability of NCM85 at high voltage, which leads to an outer shell of spinel/rock salt on the surface of the NCM85 particles that will block Li ion diffusion.^{63,64} Furthermore, upon high-potential cycling, NCM85 faces significant volume changes, which leads to crack formation within the particles and the cathode composite (i.e., contact loss at the interfaces). This contributes to the capacity fade.^{65,66}

To demonstrate full cell performance, Si was chosen as the anode material with no addition of carbon or solid electrolyte (Figure S14b). High-loading NCM85 ASSBs exhibit excellent

long-term cycling stability and maintain 80% capacity retention over ~1500 cycles (Figure 5a). Even at an ultrahigh loading of 81.3 mg_{NCM85} cm⁻², the cell can still deliver a 13.4 mAh cm⁻² discharge capacity and stable cycling over 110 cycles (Figure 5c). Due to the instability between lithiated Si and Li₆PS₅Cl, during the initial charge, some Li from NCM85 is consumed to form the interphase between Li_xSi and Li₆PS₅Cl, which leads to a low initial coulombic efficiency of ~78% (Figure 5b), and thus low NCM85 utilization. To improve this, Si was prelithiated to Li_{0.7}Si (Figure S14c), taking into account the irreversibility of NCM85 and Si during the initial cycle and required anode interface formation. The same ultrahigh-loading ASSB (81.89 mg_{NCM85} cm⁻²) with a Li_{0.7}Si anode exhibits a much higher initial coulombic efficiency of 94.5% (Figure 5e) and delivers a higher discharge areal capacity of 16.1 mAh cm⁻² (197 mAh g⁻¹, Figure 5d). Such stable cycling for ultrahigh-loading ASSBs suggests that further improvements in the energy density of ASSBs are possible. Relatively low N/P ratios (1.2 and 1.4) were used based on the theoretical capacities for the electrodes (NCM85, 200 mAh g⁻¹; Si, 3579 mAh g⁻¹; Li_{0.7}Si, 2481 mAh g⁻¹). The latter are close to the values measured in Li half-cells at a much lower applied pressure of 30 MPa (Si, 2965 mAh g⁻¹; Li_{0.7}Si, 1960 mAh g⁻¹ respectively; Figure S18), but because of the pressure difference, we prefer to use the theoretical capacity, which sets an upper bar for the N/P ratio. Due to the limitation of lithium diffusion in Si/Li_xSi anodes,⁶⁷ high current densities can lead to Li plating and hence short circuits at high areal capacity. The rate performance of high-loading ASSBs is also constrained by electronic and ionic percolation within the cathode composites;⁶⁸ thus, a more effective cathode composite design is needed.

In summary, here we report two new classes of chloride solid electrolytes, Li_{3-x}Zr_xHo_{1-x}Cl₆ and Li_{3-x}Zr_xLu_{1-x}Cl₆ (0 ≤ x ≤ 0.8), which exhibit high ionic conductivity up to 1.8 mS cm⁻¹. Utilizing high-resolution powder neutron diffraction, we identified the Li ion distribution in trigonal Li₃HoCl₆, orthorhombic-I Li₃LuCl₆, and orthorhombic-II Li_{2.4}Zr_{0.6}(Ho/Lu)_{0.4}Cl₆ and demonstrated the existence of well-connected lithium diffusion pathways rich in Li vacancies that promote lithium diffusion. Excellent chemical and electrochemical stabilities between Li_{2.6}Zr_{0.4}(Ho/Lu)_{0.6}Cl₆ and NCM85 are observed on cycling up to 4.6 V vs Li⁺/Li. Even with a 4.8 V cutoff potential, ASSBs can still maintain relatively stable cycling. A ToF-SIMS analysis revealed slightly different dominant solid electrolyte degradation mechanisms for Li_{2.6}Zr_{0.4}Ho_{0.6}Cl₆ and Li_{2.6}Zr_{0.4}Lu_{0.6}Cl₆, where the former is primarily chemical and the latter is predominantly electrochemical. Most importantly, full cells with Si or Li_{0.7}Si anodes exhibit excellent long-term cycling with high areal capacities of up to 16.3 mAh cm⁻². The prelithiated Si anode can also be generally applied for other systems, i.e., all-solid-state Li₂S batteries. These findings provide important guidance toward the development of high-voltage all-solid-state full cells.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenerylett.3c00763>.

Experimental methods, crystallographic details, Nyquist plots, electrochemical data, SEM, and mass spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Jürgen Janek – Institute of Physical Chemistry and Center for Materials Research, Justus Liebig University Giessen, 35392 Giessen, Germany; orcid.org/0000-0002-9221-4756; Email: juergen.janek@pc.jlug.de

Linda F. Nazar – Department of Chemistry and the Waterloo Institute for Nanotechnology, University of Waterloo, Waterloo, Ontario N2L 3G1, Canada; Joint Center for Energy Storage Research, Argonne National Laboratory, Lemont, Illinois 60439, United States; orcid.org/0000-0002-3314-8197; Email: lfnazar@uwaterloo.ca

Authors

Laidong Zhou – Department of Chemistry and the Waterloo Institute for Nanotechnology, University of Waterloo, Waterloo, Ontario N2L 3G1, Canada; Joint Center for Energy Storage Research, Argonne National Laboratory, Lemont, Illinois 60439, United States; orcid.org/0000-0002-8556-3296

Tongtong Zuo – Institute of Physical Chemistry and Center for Materials Research, Justus Liebig University Giessen, 35392 Giessen, Germany

Chang Li – Department of Chemistry and the Waterloo Institute for Nanotechnology, University of Waterloo, Waterloo, Ontario N2L 3G1, Canada; Joint Center for Energy Storage Research, Argonne National Laboratory, Lemont, Illinois 60439, United States; orcid.org/0000-0001-5420-3856

Qiang Zhang – Neutron Scattering Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37830, United States; orcid.org/0000-0003-0389-7039

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acsenerylett.3c00763>

Notes

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